

# MEDICAL INVENTORY MANAGEMENT PLATFORM

## MILESTONE-3

### Module Documentation

Expiry Tracking  
Low-Stock Alerts  
Notifications  
Inventory Analytics  
Report Generation

# 1. Expiry Tracking

## 1.1 Overview

The Expiry Tracking module is designed to monitor the expiry dates of medicines stored in the inventory. It helps pharmacists and staff identify medicines that are approaching their expiry date and medicines that have already expired.

## 1.2 Objectives

- Track the expiry date of every medicine.
- Identify medicines that are near their expiry date.
- Identify medicines that have already expired.
- Display the current expiry status of medicines.
- Help users take timely action on near-expiry medicines.
- Prevent expired medicines from being used or sold.

## 1.3 Key Features

- Store expiry dates along with medicine information.
- Compare medicine expiry dates with the current date.
- Categorize medicines as Valid, Near Expiry, or Expired.
- Display expiry status on the dashboard and inventory pages.
- Generate expiry-related notifications.
- Include expiry information in reports.

## 1.4 Working Process

1. The expiry date is stored when a medicine is added to the inventory.
2. The system retrieves the current date.
3. The system compares the current date with the medicine's expiry date.
4. Based on the comparison, the medicine is assigned an expiry status.
5. Near-expiry medicines are highlighted for user attention.
6. Expired medicines are identified and displayed separately.
7. The system can generate notifications for medicines approaching expiry.

## 1.5 Expected Outcome

The module provides better control over medicine expiry and reduces the risk of expired medicines remaining unnoticed in the inventory.

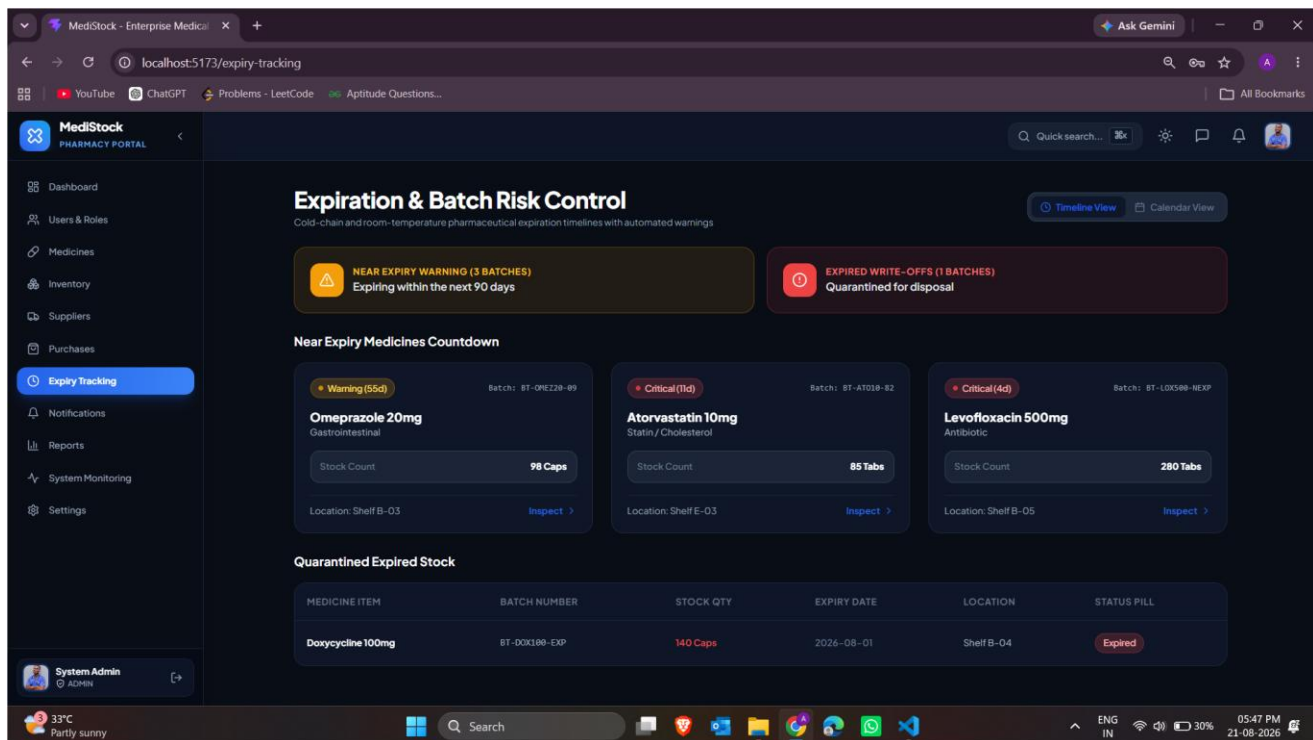


Fig : Expiry Tracking

## 2. Low-Stock Alerts

### 2.1 Overview

The Low-Stock Alerts module monitors the available quantity of medicines and identifies medicines whose stock falls below the required minimum stock level. It also identifies medicines that have completely run out of stock and provides alerts so that users can take necessary replenishment actions.

### 2.2 Objectives

- Monitor medicine quantities continuously.
- Identify medicines below the minimum stock level.
- Detect out-of-stock medicines.
- Display low-stock information on the dashboard.
- Generate alerts for low-stock medicines.
- Support timely stock replenishment.

### 2.3 Stock Categories

| Stock Condition | Description                                                        |
|-----------------|--------------------------------------------------------------------|
| Normal Stock    | Quantity is above the minimum stock level                          |
| Low Stock       | Quantity is greater than 0 but at or below the minimum stock level |
| Out of Stock    | Quantity is 0                                                      |

Example: If the minimum stock level is 10, a quantity greater than 10 is Normal Stock, a quantity from 1 to 10 is Low Stock, and a quantity of 0 is Out of Stock.

## 2.4 Key Features

- Maintain the quantity of each medicine.
- Store a minimum stock level.
- Automatically check stock quantities.
- Identify low-stock medicines.
- Identify out-of-stock medicines.
- Display stock status on dashboards.
- Generate low-stock notifications.
- Support inventory replenishment decisions.

## 2.5 Working Process

8. The system stores the current quantity of each medicine.
9. A minimum stock level is defined for the medicine.
10. The backend compares the available quantity with the minimum stock level.
11. If the quantity is below or equal to the minimum level, the medicine is marked as low stock.
12. If the quantity becomes zero, it is marked as out of stock.
13. The system generates an appropriate notification.
14. The dashboard displays the updated stock information.

## 2.6 Expected Outcome

The module helps prevent unexpected stock shortages and ensures that important medicines can be replenished before they become unavailable.

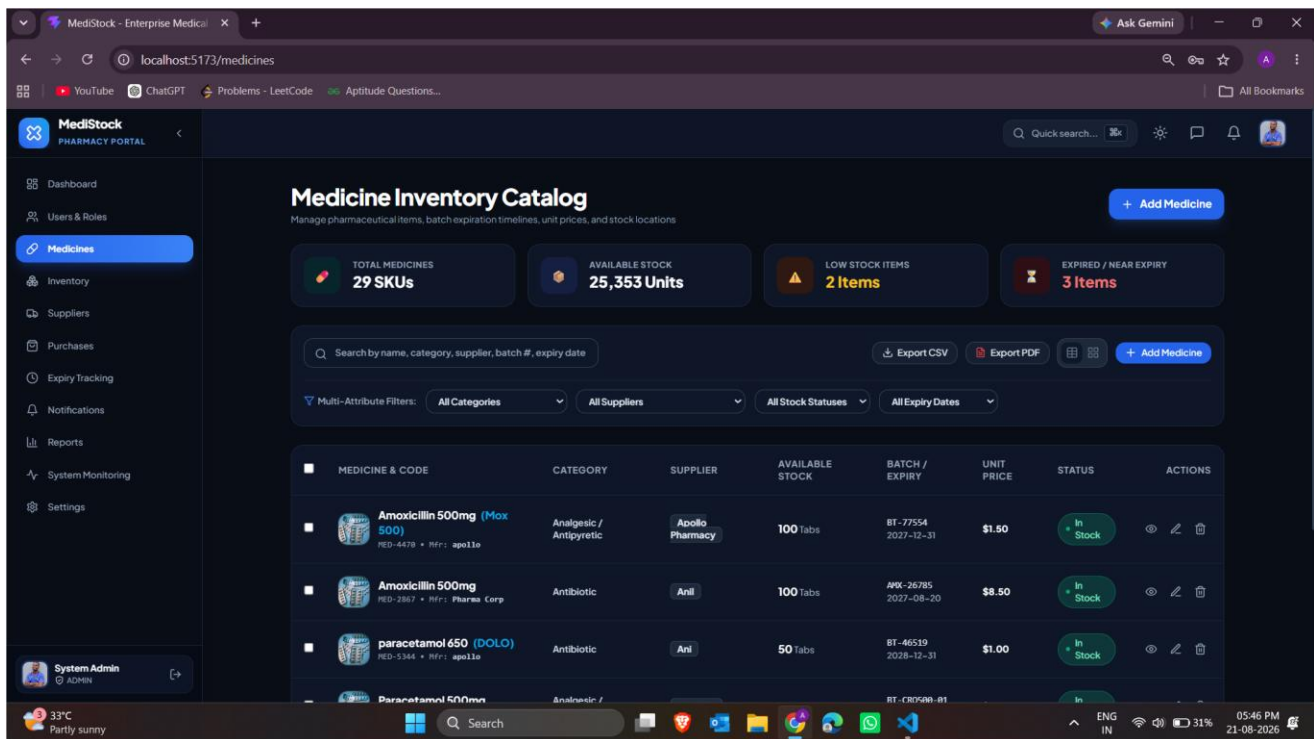


Fig : Low Stock

## 3. Notifications

### 3.1 Overview

The Notifications module provides users with important and timely information about inventory conditions. It automatically generates notifications based on events such as low stock and approaching medicine expiry.

### 3.2 Objectives

- Notify users about low-stock medicines.
- Notify users about expired or near-expiry medicines.
- Provide system-generated inventory alerts.
- Display notifications according to the user's role.
- Maintain notification read/unread status.

### 3.3 Notification Types

| Notification Type | Purpose                                                        |
|-------------------|----------------------------------------------------------------|
| LOW_STOCK         | Indicates that medicine quantity is below the required level   |
| EXPIRY            | Indicates that a medicine is approaching or has reached expiry |
| STOCK_UPDATE      | Indicates an inventory stock change                            |
| SALE              | Indicates a medicine sale or inventory transaction             |

### 3.4 Key Features

- Automatic notification generation.
- Low-stock notifications.
- Expiry notifications.
- Role-based notifications.
- Read/unread notification status.
- Notification count display.
- Notification list on the dashboard.
- Notification history.

### 3.5 Working Process

15. The backend monitors inventory information.
16. Stock levels and expiry dates are checked.
17. When a predefined condition is satisfied, the system creates a notification.
18. The notification is stored in the notification database.
19. The appropriate user retrieves the notification through the notification API.
20. The dashboard displays the notification.
21. Users can view and mark notifications as read.

### 3.6 Examples

Low-stock notification: "Paracetamol stock is low. Available quantity: 5."

Expiry notification: "Amoxicillin is approaching its expiry date."

### 3.7 Expected Outcome

The notification system ensures that users are informed about important inventory events without having to manually monitor the complete inventory.

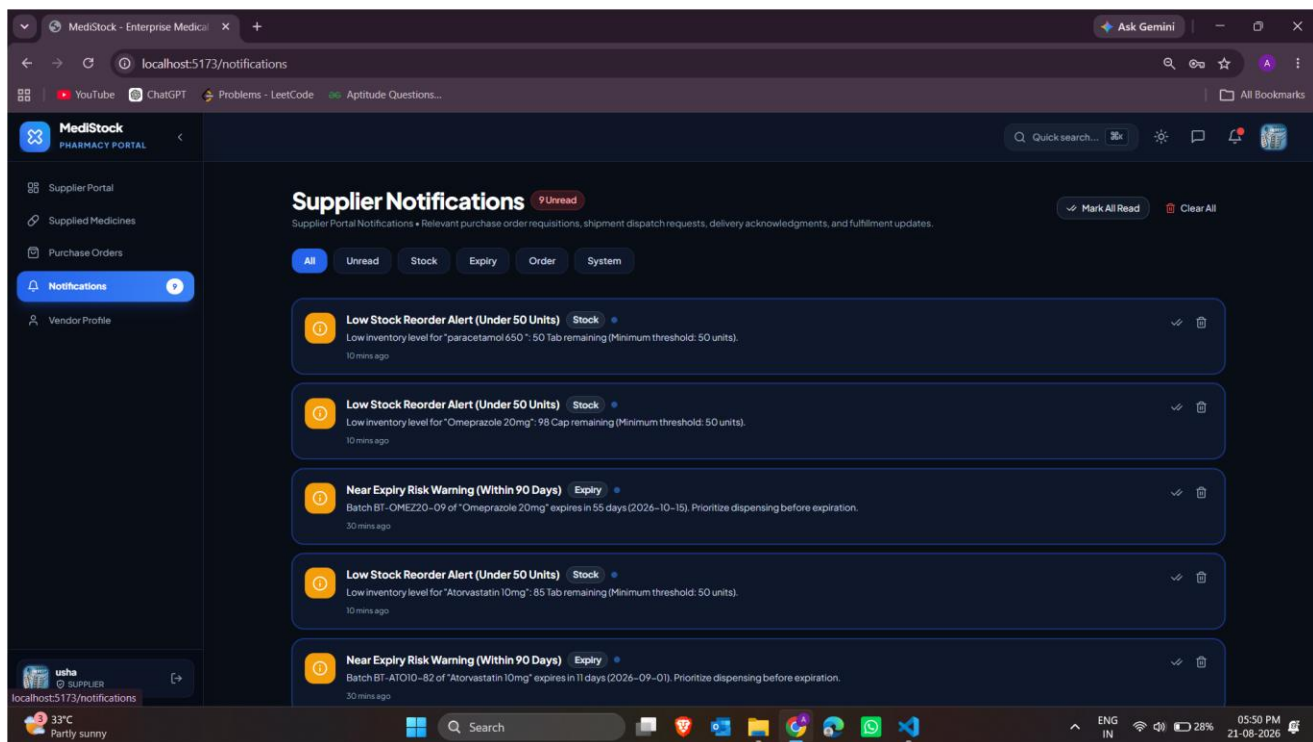


Fig : Supplier Notifications

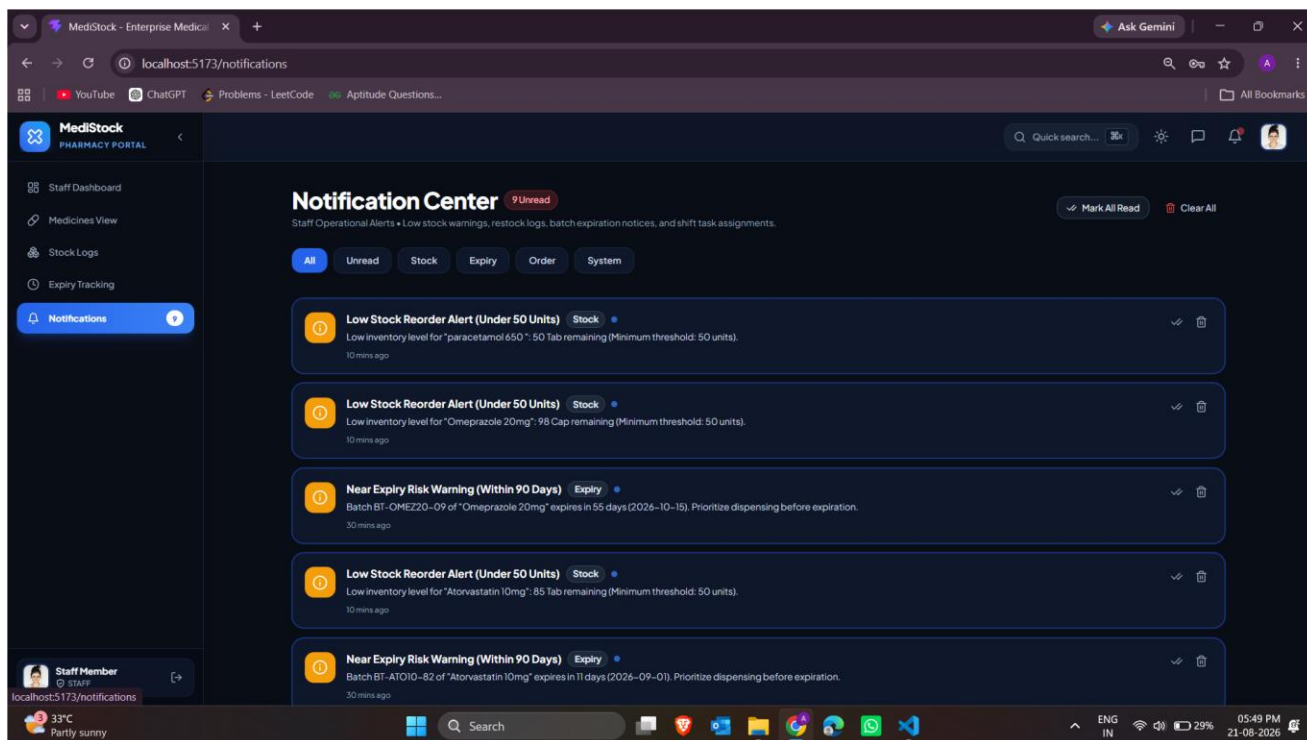


Fig : Staff Notifications

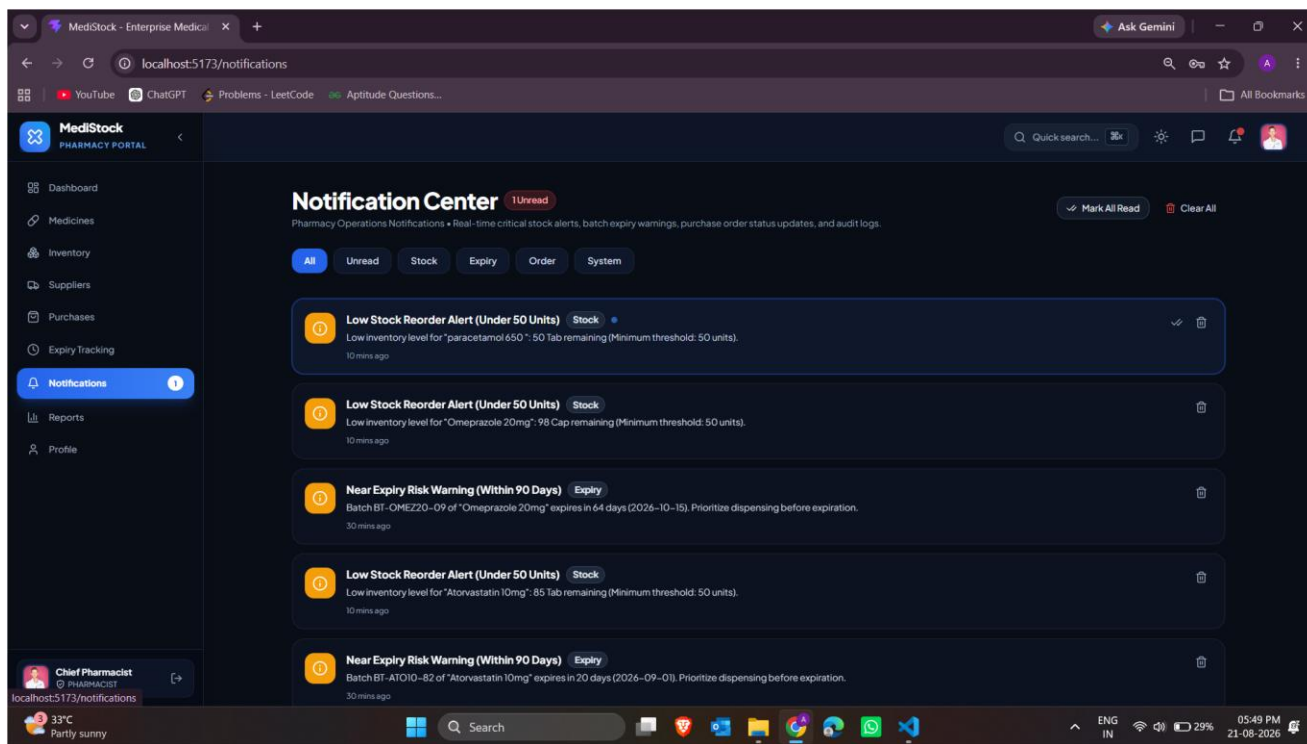


Fig : Pharmacist Notifications

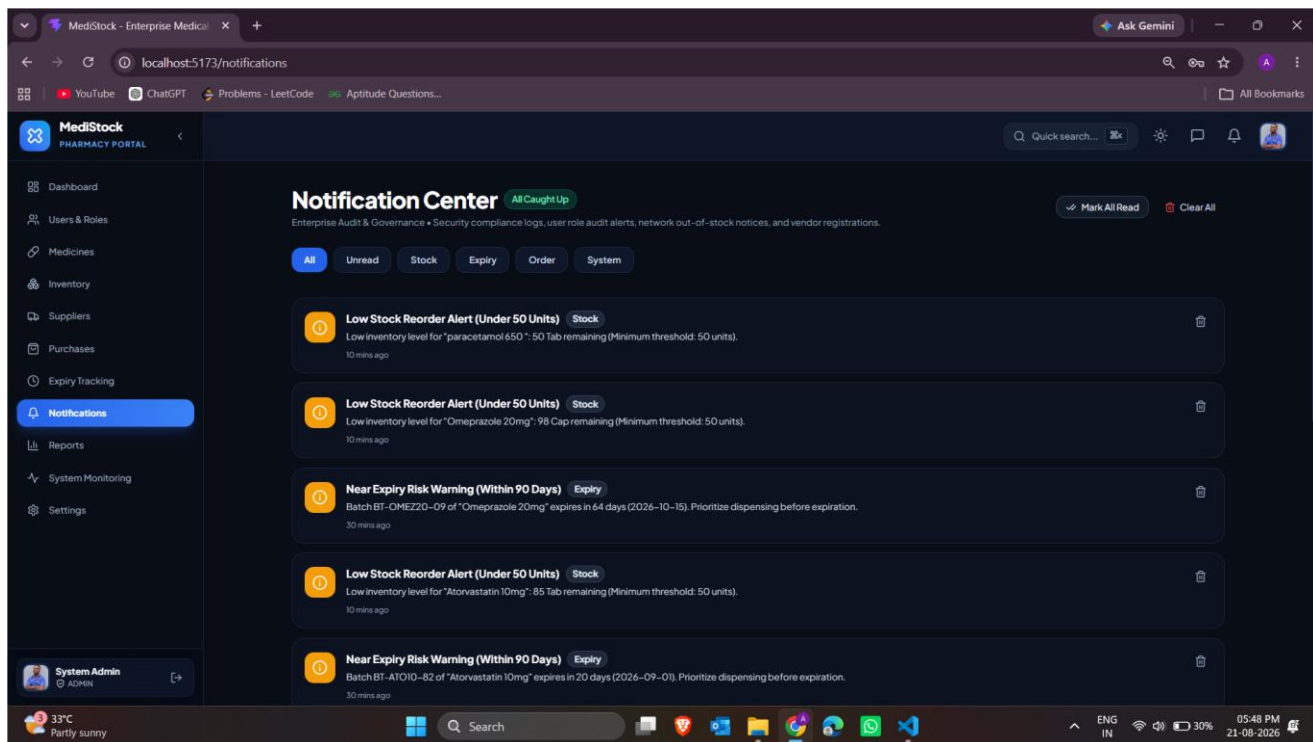


Fig : Admin Notifications

## 4. Inventory Analytics

### 4.1 Overview

The Inventory Analytics module provides summarized information about the overall inventory. Backend APIs are used to calculate important inventory metrics, and the results are displayed on the dashboard.

### 4.2 Objectives

- Calculate total medicines in the inventory.
- Calculate total available stock.
- Identify low-stock medicines.
- Identify out-of-stock medicines.
- Identify expiring medicines.
- Provide summarized inventory information.
- Display analytics on the dashboard.

### 4.3 Analytics Metrics

| Metric          | Description                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------|
| Total Medicines | Total number of different medicines maintained in the inventory.                                 |
| Available Stock | Total quantity of medicines currently available.                                                 |
| Low-Stock Items | Number of medicines whose quantity is at or below the minimum stock level but greater than zero. |

|                    |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| Out-of-Stock Items | Number of medicines whose available quantity is zero.                                 |
| Expiring Medicines | Medicines approaching their expiry date according to the configured expiry threshold. |

#### 4.4 Backend Analytics API

Example endpoint:

##### GET /api/analytics

Example response:

```
{
  "totalMedicines": 100,
  "availableStock": 2500,
  "lowStockItems": 12,
  "outOfStockItems": 5,
  "expiringMedicines": 8
}
```

#### 4.5 Working Process

22. The frontend requests analytics information from the backend.
23. The backend retrieves medicine and inventory data from the database.
24. Required calculations are performed.
25. The calculated results are returned through REST APIs.
26. The frontend receives the analytics data.
27. The dashboard displays the results using cards, tables, or other visual components.

#### 4.6 Dashboard Representation

- Total Medicines
- Available Stock
- Low Stock
- Out of Stock
- Expiring Medicines

#### 4.7 Expected Outcome

The analytics module provides a centralized view of inventory performance and helps users quickly identify important stock and expiry-related issues.

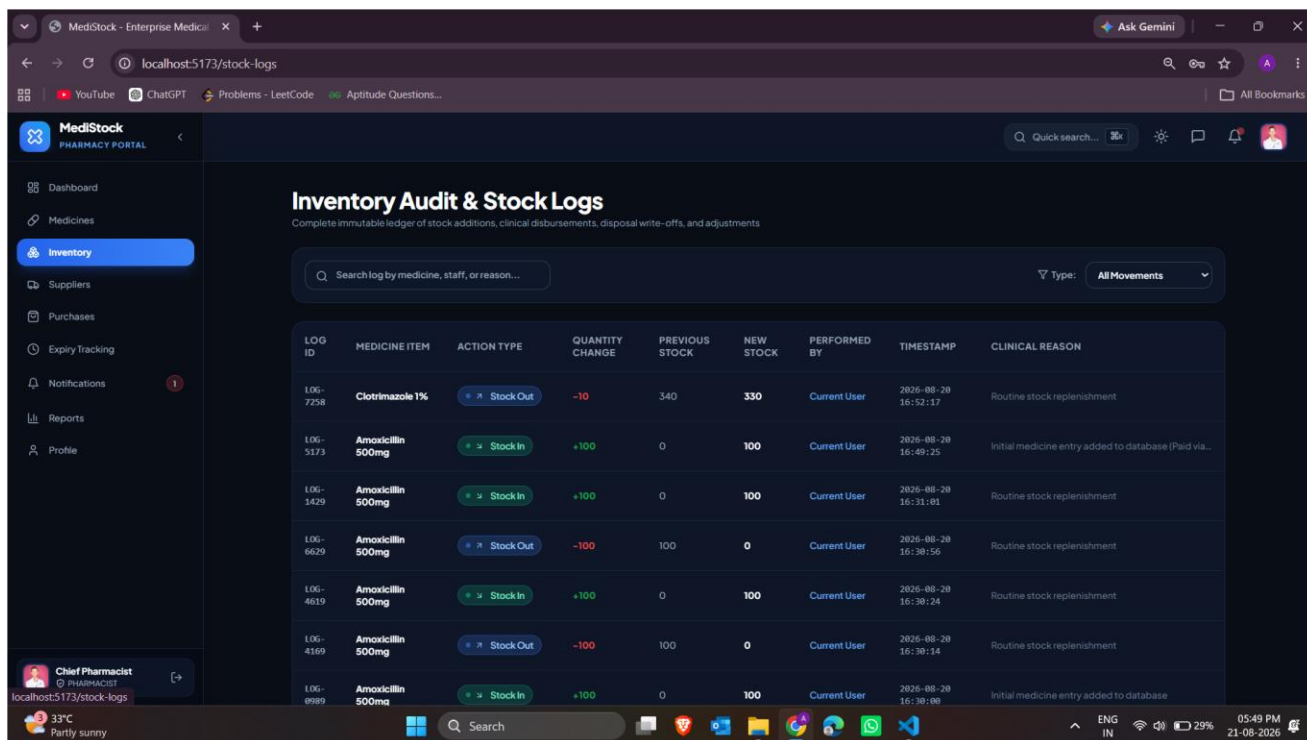


Fig : Pharmacist Inventory

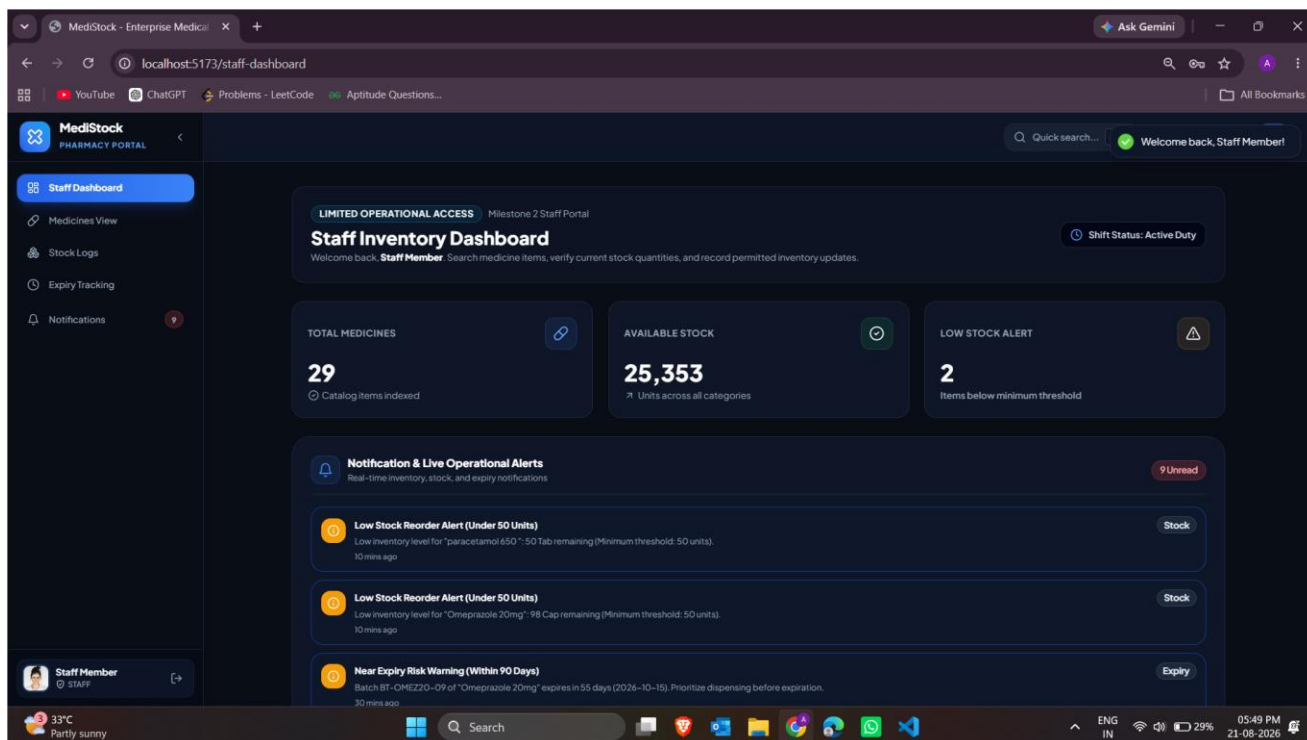


Fig : Staff Inventory

**Assigned Medicine Inventory Purchases**  
View inventory purchases and stock allocated directly to your dispensary unit

Search by PO ID, medicine, pharmacist, supplie

All Suppliers All Pharmacists All Statuses 11 DESC

| PO ID / PHARMACIST | MEDICINE NAME                     | SUPPLIER                 | QUANTITY & TOTAL           | PURCHASE DATE | BATCH / EXPIRY                     | STATUS    | ACTIONS   |
|--------------------|-----------------------------------|--------------------------|----------------------------|---------------|------------------------------------|-----------|-----------|
| PO-8801<br>JohnDoe | Paracetamol 650mg (Dolo 650)      | MedPlus                  | \$25,000.00<br>10000 units | Aug 10, 2026  | BT-DOL0650-99<br>Exp: Aug 15, 2027 | Approved  |           |
| PO-8805<br>JohnDoe | Cetirizine 10mg                   | Img / Tata Img           | \$14,000.00<br>4000 units  | Aug 9, 2026   | BT-CET10-88<br>Exp: Jan 18, 2028   | Completed | Restocked |
| PO-8810<br>JohnDoe | Ibuprofen 400mg (Brufen 400)      | Abbott                   | \$12,000.00<br>8000 units  | Aug 7, 2026   | BT-IBU400-55<br>Exp: May 15, 2028  | Completed | Restocked |
| PO-8803<br>JohnDoe | Azithromycin 500mg (Azithral 500) | PharmEasy                | \$44,000.00<br>2000 units  | Aug 5, 2026   | BT-AZ1500-04<br>Exp: Sep 1, 2027   | Completed | Restocked |
| PO-8804<br>JohnDoe | Paracetamol 500mg (Crocin)        | MedPlus                  | \$24,000.00<br>12000 units | Jul 28, 2026  | BT-CRO500-01<br>Exp: Oct 1, 2027   | Completed | Restocked |
| PO-8808<br>JohnDoe | Metformin 500mg (Glycomet 500)    | Dr. Reddy's Laboratories | \$45,000.00<br>15000 units | Jul 25, 2026  | BT-GLY500-21<br>Exp: Jan 28, 2028  | Completed | Restocked |

Showing 1 to 6 of 6 Purchases

Page 1 of 1

Fig : Assigned Medicine Inventory

**Medicine Inventory Catalog**  
Manage pharmaceutical items, batch expiration timelines, unit prices, and stock locations

29 SKUs AVAILABLE STOCK 25,353 Units LOW STOCK ITEMS 2 Items EXPIRED / NEAR EXPIRY 3 Items

Search by name, category, supplier, batch #, expiry date

Export CSV Export PDF Add Medicine

Multi-Attribute Filters: All Categories All Suppliers All Stock Statuses All Expiry Dates

| MEDICINE & CODE                                       | CATEGORY                | SUPPLIER        | AVAILABLE STOCK | BATCH / EXPIRY          | UNIT PRICE | STATUS   | ACTIONS |
|-------------------------------------------------------|-------------------------|-----------------|-----------------|-------------------------|------------|----------|---------|
| Amoxicillin 500mg (Mox 500)<br>RED-4470 • Ref: apollo | Analgesic / Antipyretic | Apollo Pharmacy | 100 Tabs        | BT-77554<br>2027-12-31  | \$1.50     | In Stock |         |
| Amoxicillin 500mg<br>RED-2867 • Ref: Pharma Corp      | Antibiotic              | Anil            | 100 Tabs        | AWX-26785<br>2027-08-20 | \$8.50     | In Stock |         |
| paracetamol 650 (DOLO)<br>RED-3344 • Ref: apollo      | Antibiotic              | Ani             | 50 Tabs         | BT-46519<br>2028-12-31  | \$1.00     | In Stock |         |

Fig : Admin Inventory

## 5. Report Generation

### 5.1 Overview

The Report Generation module allows users to generate reports related to medicine inventory and expiry information. These reports provide structured information that can be used for monitoring, analysis, and record keeping.

### 5.2 Objectives

- Generate inventory reports.
- Generate expiry reports.
- Provide structured inventory information.
- Support report export functionality.
- Help users maintain inventory records.
- Assist administrators in analyzing inventory conditions.

### 5.3 Types of Reports

#### Inventory Report

- Medicine name
- Category
- Manufacturer
- Batch number
- Available quantity
- Minimum stock level
- Price
- Supplier
- Stock status

#### Expiry Report

- Medicine name
- Batch number

- Expiry date
- Current status
- Days remaining
- Available quantity

## 5.4 Report Generation Process

28. The user selects the required report type.
29. The frontend sends a request to the backend.
30. The backend retrieves the required inventory information.
31. The system processes and organizes the data.
32. The report is generated.
33. The user can view or export the report.

## 5.5 Export Functionality

The system can provide export options such as:

- PDF
- CSV
- Excel

## 5.6 Expected Outcome

The Report Generation module provides users with organized inventory and expiry information and makes it easier to maintain records and perform inventory analysis.



Fig : Admin Report



Fig : Pharmacist Report

## 6. Overall Module Integration

The five modules work together to provide an effective medicine inventory management system. Inventory and medicine data are processed by the backend, which performs expiry and stock checks, generates analytics and notifications, and supplies the dashboard and reporting functions.

| Module              | Main Responsibility                                 |
|---------------------|-----------------------------------------------------|
| Expiry Tracking     | Monitors near-expiry and expired medicines.         |
| Low-Stock Alerts    | Identifies low-stock and out-of-stock medicines.    |
| Notifications       | Provides automatic inventory alerts.                |
| Inventory Analytics | Calculates and displays inventory insights.         |
| Report Generation   | Generates and exports inventory and expiry reports. |

## 7. Benefits

34. Better monitoring of medicine expiry dates.
35. Early identification of near-expiry medicines.
36. Prevention of expired medicines being used or sold.
37. Automatic identification of low-stock medicines.
38. Detection of out-of-stock medicines.
39. Automatic inventory notifications.
40. Centralized inventory analytics.
41. Better visibility of inventory conditions through dashboards.
42. Easy generation of inventory and expiry reports.

43. Improved decision-making for medicine purchasing and stock management.

## 8. Conclusion

The Expiry Tracking, Low-Stock Alerts, Notifications, Inventory Analytics, and Report Generation modules form important components of the Medical Inventory Management Platform. Together, they automate important inventory monitoring activities and provide users with timely information about stock availability and medicine expiry.

By integrating these modules with the React frontend, Spring Boot backend, and database, the system provides a centralized and efficient solution for managing medicine inventory, reducing stock-related problems, improving expiry management, and supporting better inventory decisions.